

Optimization: Critical Points & Lagrange Multipliers



Finding critical points

- 1 Find the critical point(s) of $f(x, y) = x^2 + 3y^2 - 4x + 6y$.
 - 2 Find the critical point(s) of $f(x, y) = xy - x^2 - y^2$.
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The Hessian and the Second Derivative Test

- 3 Classify the critical point $(2, -1)$ of $f(x, y) = x^2 + 3y^2 - 4x + 6y$ using $D = f_{xx}f_{yy} - (f_{xy})^2$.
 - 4 Classify the critical point $(0, 0)$ of $f(x, y) = xy - x^2 - y^2$.
 - 5 Classify the critical point of $f(x, y) = x^2 - y^2$.
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Lagrange multipliers: setting up the system

- 6 Set up the Lagrange multiplier equations to optimize $f(x, y) = xy$ subject to $g(x, y) = x + y - 10 = 0$.
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Solving Lagrange multiplier problems

- 7 Maximize $f(x, y) = xy$ subject to $x + y = 10$.
 - 8 Minimize $f(x, y) = x^2 + y^2$ subject to $x + y = 4$.
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A geometric Lagrange multiplier application

- 9 Find the point on the line $x + y = 6$ closest to the origin, using Lagrange multipliers on $f(x, y) = x^2 + y^2$.
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More critical point practice

- 10 Find the critical point(s) of $f(x, y) = x^2 + y^2 - 6x + 2y + 5$.
 - 11 Find the critical point(s) of $f(x, y) = x^3 - 3xy + y^3$.
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More Hessian classification practice

- 12 Classify the critical point $(3, -1)$ of $f(x, y) = x^2 + y^2 - 6x + 2y + 5$.
 - 13 Classify the critical points $(0, 0)$ and $(1, 1)$ of $f(x, y) = x^3 - 3xy + y^3$ found above.
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An applied unconstrained optimization problem

- 14 A firm's profit from producing x, y units of two goods is $P(x, y) = 100x + 120y - 2x^2 - 3y^2 - 2xy$. Find the production levels that maximize profit.
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Lagrange multipliers with a circular constraint

- 15 Set up the Lagrange multiplier equations to optimize $f(x, y) = x + y$ subject to $x^2 + y^2 = 8$.
- 16 Find the maximum and minimum values of $f(x, y) = x + y$ subject to $x^2 + y^2 = 8$.
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An applied Lagrange multiplier problem

- 17 A farmer has 100 m of fencing to enclose a rectangular field along a straight river (no fence needed along the river side). If x is the side parallel to the river and y each perpendicular side, maximize the area $A = xy$ subject to $x + 2y = 100$.
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One more critical point and classification

- 18 Find and classify the critical point(s) of $f(x, y) = x^2 + xy + y^2$.
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A closing Lagrange multiplier problem

- 19 Minimize $f(x, y) = x^2 + y^2$ subject to $xy = 1$ (with $x, y > 0$).
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A final critical-point classification

- 20 Find and classify the critical point(s) of $f(x, y) = x^2y - 6y^2 - 3x^2$.
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Global extrema on a closed region

- 21 Find the maximum and minimum of $f(x, y) = x^2 + y^2$ on the closed disk $x^2 + y^2 \leq 4$ (check both the interior critical point and the boundary).
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One last constrained-optimization setup

- 22 A box with a square base of side x and height y has volume $V = x^2y = 32$. Set up the Lagrange equations to minimize the surface area $S = 2x^2 + 4xy$ subject to this constraint.
- 23 Solve the box problem above: find x and y that minimize $S = 2x^2 + 4xy$ subject to $x^2y = 32$.
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Second derivative test practice, continued

24 Find and classify the critical point(s) of $f(x, y) = 4 + x^3 + y^3 - 3xy$.

A Lagrange multiplier problem in three variables

25 Set up the Lagrange equations to minimize $f(x, y, z) = x^2 + y^2 + z^2$ subject to $x + y + z = 6$.

26 Solve the problem above.